

Week 11: Decoder-Only LLMs

Sebastián Vallejo Vera ¹

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¹ Dept. of Political Science – Western University

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Transformers in Decoders

Generative Large Language Models

- Large Language Models (LLM) are pre-trained decoders. This means models that have been pre-trained using the Transformers architecture to generate new text from a given prompt.
- In their most basic understanding, LLMs excel at next word prediction. So good, that people (erroneously, in my view) confound LLM responses with human-like behavior.
- The SOTA LLM as of July 2026 are: GPT-5something (OpenAI), Llama4 something (Meta), DeepSeek RSomething, Gemini 3something, Claude 4something, Kimi K3 (?)... it's an industry reminiscent of dot.com bubble (again, in my view). By the time you re-read this, it will be outdated.
- The structure is pretty straightforward: (pre-prompt) $\Rightarrow$ prompt $\Rightarrow$ LLM output $\Rightarrow$ prompt $\Rightarrow$ LLM output etc.

Why Are LLMs So Good, Though? i

- LLMs are trained using gazillions of text data they stole scraped from the internet. For example, GPT-3 (outdated model) was trained using 500 billion tokens and 175 billion parameters.
- Many of these models are proprietary, so we have few clues as to why they are so good. Even the ones that claim to be open-source (like Llama), are not really open-source. The ethical standing of companies developing LLMs is, in my view, tenuous at best (but we still use these models, so its a cycle of hypocrisy, as is the norm in this late stage of capitalism).
- More importantly, even if there is a truly open-source LLM out there, it would be close to impossible for you or me to actually peruse their code and figure out what is making them work.

Why Are LLMs So Good, Though? ii

- Beyond the fact that they have troves of data which they transform (into tensor networks) and spew back at us (which is bound to make things pretty good at mimicking the people producing that data), one mechanism that seems to be at work is **in-context learning**.
 - During pre-training, LLMs develop certain 'skills', among them pattern recognition. When prompted, LLMs rapidly recognize the required task–i.e., in-context learning– and adapt their output accordingly.
 - Why this happens is still unclear.

- LLMs have been increasingly used in the social sciences.
- Goal: generate data from text (next token prediction); measure some latent characteristic of a set of texts.
 - APPLIED: latent dimensions, i.e., ideology (Kato and Cochrane, 2025; Wu et al., 2023); annotators (Timoneda and Vallejo Vera, 2025); synthetic data (Li et al., 2023); and others...
 - LIMITATIONS: biases from party cues (Vallejo Vera and Driggers, 2025); bias from language (Walker and Timoneda, 2024).
 - META: effect of LLMs on respondents.

Applications and Limitation of LLMs

Applications of LLM

- With LLMs, what researchers can do is really limited by creativity. I will provide the basic code structure to access LLMs through an API, and obtaining and tidying data, but most of the responses will come from prompts. So... prompt engineering? (I do not like this term or this practice but it's a thing).
- There are some approaches that improve responses, and there are some best practices to consider when using LLMs. I will cover them both in the slides and provide examples of how to implement these in the code.

To the **code!**

Classification Tasks with LLM i

- One of the most common applications of LLMs is text classification (annotation).
 - Ask for a list of topics from sets of texts (unsupervised).
 - Ask for a specific classification from a set of options (supervised).
- Approaches to classifying text:
 - Zero-shot learning: Only instructions in the prompt, no examples.
 - Few-shot learning: Instructions + examples (between 5 and 10).
 - Few-shot with chain-of-thought (CoT) reasoning: Ask model to classify example + ask reasoning behind classification (in steps if necessary) + add (example text + classification + reasoning) to the prompt.
- Few-shot CoT improves performance but has a major problem: examples do not generalize well to unseen new texts, gains are limited, and more examples do not improve performance (overfitting).

- (More self-promotion coming up) We (Timoneda and Vallejo Vera, 2025, 2026) propose a new approach called **memory learning**:
 - We allow the model to learn from its own previous classifications, which should improve performance in supervised classification.
 - Similar to how human annotators code data – we learn as we go.
 - More technical: Models already excel with 'in-context learning' and use 'contextual memory', so we can maximize learning by adding more memory

To the **code!**

Best Practices when Using LLM i

(Note that these are mostly from my own research and application)

- Since LLMs are non-deterministic in their answers, you should run multiple iterations of each annotation task. Internal consistency will vary across models and hyperparameters used.
- Best practices when using LLMs as annotators include multiple iterations of the task, reporting on all the output from LLMs, and clearly explaining the adjudication strategy employed. This increases transparency and allows for replication.
- Still, the constantly-changing LLM environment makes replication complicated and, with time, impossible. Researchers should clearly report the version of the model used and, if possible, the platform on which it was deployed (or if it was locally deployed).

- My final recommendation goes back to Grimmer and Stewart (2013) key principle of text analysis more generally: “validate, validate, validate.”

Limitations when Using LLM i

(Note that these are also mostly from my own research and application)

- Researchers should take into account that LLMs are not created in a vacuum. They are the partial product of **human-generated** data, that is embedded with human biases.
- Thus, results will mimic human biases, **but in a very LLM way** (see Vallejo Vera and Driggers, 2025).
- Fine-tuning a Transformers-based model will still be more accurate for text classification than using a LLM (because we take advantage of the full encoder structure, and fully update weights during fine-tuning).

Limitations when Using LLM ii

- Models are trained in high-resource languages... mostly English. They are great at English language tasks. But most of the world is not in English.
- Finally, the broader class/core-periphery/empire considerations of access to knowledge and resources are applied to LLM (and computational text-analysis more broadly).

Helpful Information and Discussions about LLMs

Technical References to LLM

Here are some technical references about LLMs. Mostly, references about different aspects of the inner workings of LLMs that might be helpful for your own research.

- PROMPT CACHING: What is prompt caching? There are many matrix multiplications going on in the Transformers architecture, some of which are being duplicated. Not duplicating them saves space (and money). Here is an excellent explainer (with amazing visualizations):

<https://ngrok.com/blog/prompt-caching>.

- LLM VISUALIZER: Another pretty cool visualizer of how a LLM works: *<https://bbycroft.net/llm>*.
- FINE-TUNE A LLM (GPT2): Would you like to train (actually fine-tune) your own LLM on the style of a particular author, person, or genre? Here is the code to do that:

<https://colab.research.google.com/drive/15wS338BwTx8ibizVv0JFSeJ7l7pESUUJ?usp=sharing>.

Ethical Concerns with LLM

LLMs come with a tonne of ethical concerns (many of which, by now, I hope I have already mentioned during lectures). Here are a few more to keep in mind.

- THE PROVENANCE PROBLEM: When using LLMs to draft any piece of (scholarly) writing, we might be using ideas of other scholars, unbeknownst to us. This is a provenance problem: <https://www.nature.com/articles/s42256-025-01159-8>. (The irony is that this paper is behind a paywall).
- ALGORITHMS ARE BIASED: Learn about algorithm bias and how to identify it: <https://www.criticalracedigitalstudies.com/peoplesguide>.

What are the consequence/changes that come from the interactions with a new technology in terms of capacity, skills, and behavior? AI is not the first 'disruptive' technology, so there is good research on the consequences of human-technology interaction that could be adapted to AI.

- MORAL DESKILLING AND UPSKILLING: A fantastic paper on moral deskilling and upskilling: <https://link.springer.com/article/10.1007/s13347-014-0156-9>.

Questions? Comments??

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